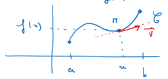
Représentations graphiques

2D

1) $f : [a,b] \longrightarrow \mathbb{R}$, $x \longmapsto f(x)$



$\mathcal{C} = \{(x,y) \in \mathbb{R}^2,\ y = f(x),\ x \in [a,b]\}$.

$M = (x, f(x))$, $\vec{v} = \left(\frac{d}{dx}(x), \frac{d}{dx} f(x)\right)$

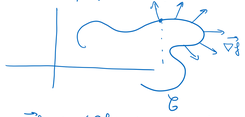
$\vec{v} = (1, f'(x))$.

2) Définition implicite des courbes.

$\mathcal{C} = \{(x,y) \in \mathbb{R}^2,\ f(x,y) = 0\}$.

Ex : $C_1 = \{(x,y) \in \mathbb{R}^2,\ x^2+y^2-1=0\}$

$C_2 = \{(x,y) \in \mathbb{R}^2,\ x+y^2-1=0\}$, ...



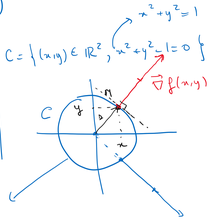
$\vec{\nabla} f(x,y) = \left(\frac{\partial f}{\partial x}(x,y), \frac{\partial f}{\partial y}(x,y)\right)$

Ex $f(x,y) = x^2+y^2-1$

$\frac{\partial f}{\partial x}(x,y) = 2x$

$\frac{\partial f}{\partial y}(x,y) = 2y$

$C = \{(x,y) \in \mathbb{R}^2,\ x^2+y^2-1=0\}$



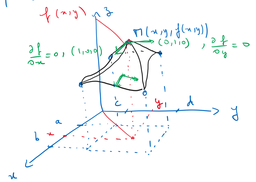
$\vec{\nabla} f(x,y) \perp$ à $\mathcal{C}$ en $M(x,y)$, avec $f(x,y) = 0$.

3D

1) En fonctionnel :

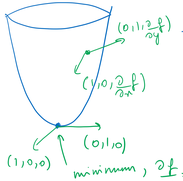
$f : D \subset \mathbb{R}^2 \longrightarrow \mathbb{R}$, $D = [a,b] \times [c,d]$.
$(x,y) \longmapsto f(x,y)$

$\mathcal{S} = \{(x,y,z) \in \mathbb{R}^3,\ z = f(x,y),\ (x,y) \in D\}$.



En M on calcule $\vec{V}_x = \left(\frac{\partial}{\partial x}(x), \frac{\partial}{\partial x}(y), \frac{\partial}{\partial x} f(x,y)\right)$

$= \left(1, 0, \frac{\partial}{\partial x} f(x,y)\right)$



$\vec{V}_y = \left(\frac{\partial}{\partial y}(x), \frac{\partial}{\partial y}(y), \frac{\partial}{\partial y} f(x,y)\right)$

$= \left(0, 1, \frac{\partial}{\partial y} f(x,y)\right)$

minimum, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$.

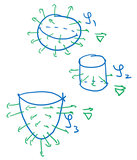
2) Définition implicite

$\mathcal{S} = \{(x,y,z) \in \mathbb{R}^3,\ f(x,y,z) = 0\}$

Ex : $\mathcal{S}_1 = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2+z^2-1=0\}$

$\mathcal{S}_2 = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2-1=0\}$

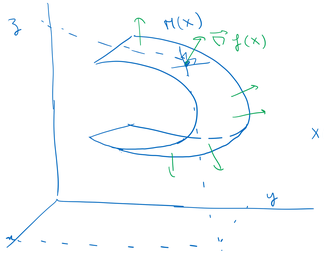
$\mathcal{S}_3 = \{(x,y,z) \in \mathbb{R}^3,\ x^2+y^2-z=0\}$

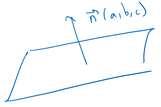


$X = (x,y,z)$

$\vec{\nabla} f(X) = \left(\frac{\partial}{\partial x} f(X), \frac{\partial f}{\partial y}(X), \frac{\partial f}{\partial z}(X)\right)$

$\vec{\nabla} f(X) \perp$ à $\mathcal{S}$ en $M(x,y,z)$ avec $f(x,y,z) = 0$.





$\underbrace{ax+by+cz+d}_{f(x,y,z)} = 0$, $X = (x,y,z)$

$\vec{\nabla} f(X) = \left(\frac{\partial}{\partial x} f(X), \frac{\partial f}{\partial y}(X), \frac{\partial}{\partial z} f(X)\right)$

$= (a,b,c) = \vec{n}$.